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GeoPops demonstration: An open-source, adaptable framework for agent-based modeling on synthetic populations

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GeoPops

- Open-source package for generating geographically and demographically realistic synthetic populations for any US Census location using publicly available data
- Same methodology as GREASYPOP-CO (One Health Trust) but wrapped in convenient Python package that can be pip installed
- Automatic data downloading
- Compatibility with Census data beyond 2019 (in development)
- Compatibility with agent-based modeling software Starsim (IDM)

github.com/ACCIDDA/GeoPops

pypi.org/project/geopops

MIDAS Notebook Tutorial

Demonstration of framework

1. Make a population of Howard County, MD
2. Run SIR Starsim model of generic upper respiratory infection
3. Incorporate prevalence-dependent "school closures"
4. Track outcomes by age and Census Block Group (CBG)

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

Output: Agents and networks

Agents
- Age
- Gender
- Race/ethnicity
- Worker status
- Student status
- School grade
- Commuter status
- Income category
- Workplace category
- Home CBG
- School CBG if applicable
- Work CBG if applicable
- GQ CBG if applicable

School
- List of students for each
- Size of grade levels
- Number of teachers
- CBG location

Household
- List of agents for each
- CBG location

Workplace
- Number of workers
- CBG if inside population area

Group quarters (GQ)
- Number of residents
- Number of staff
- CBG location

Methods: Generating agents and households

Public Use Microdata Samples (PUMS)

Combinatorial optimization (CO)

- Subset of census survey responses
- Compositional details, e.g., the number of 5-member households in a CBG with school-aged children

American Community Survey (ACS)

- Demographic data at fine geographic scale
- Limited to marginal counts, e.g., the number of households in a CBG with 5 members and the number of households in a CBG with school-aged children

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698

Methods: Assigning agents to schools and workplaces

	Data source	Data	Methods
Schools	National Center for Education Statistics (NCES)	Student and staff counts for each grade level in public schools	• Students assigned to closest school that offered required grade • Teachers assigned to school by selecting teachers from nearby CBGs
Workplaces	Longitudinal Employer-Household Dynamics (LEHD)	Origin-destination and workplace characteristics	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers

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Methods: Generating Contacts

Layer	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: α=0.9 • Mean degree: K=12 • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: α=0.9 • Mean degree: K=8 • Blocks are income levels (<40k or ≥ 40k)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: K=12 • Rewiring probability: β=0.25

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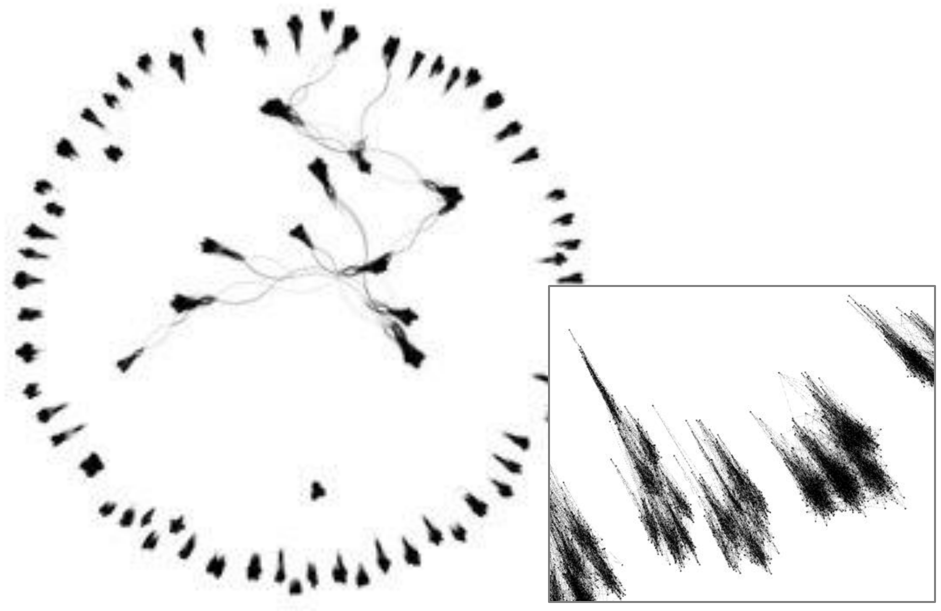
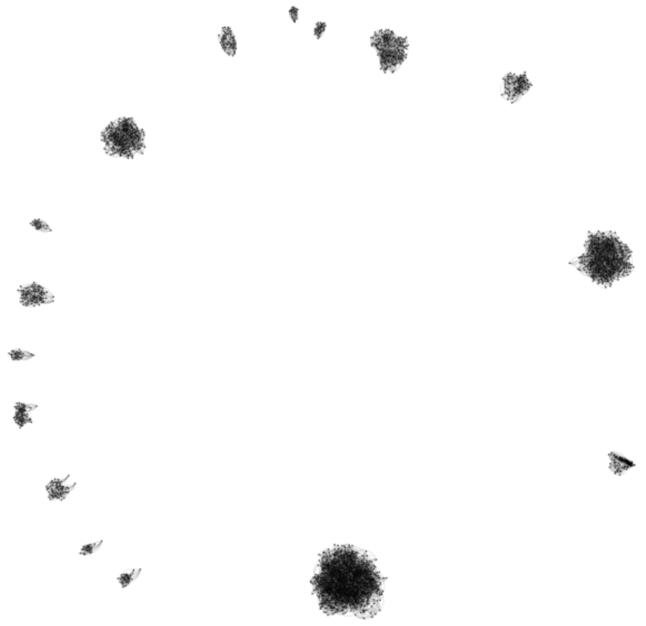
Network visualizations for Howard County, MD



School



Group Quarters

GeoPops with Starsim

Pass these elements into Starsim simulation

SIR model
- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

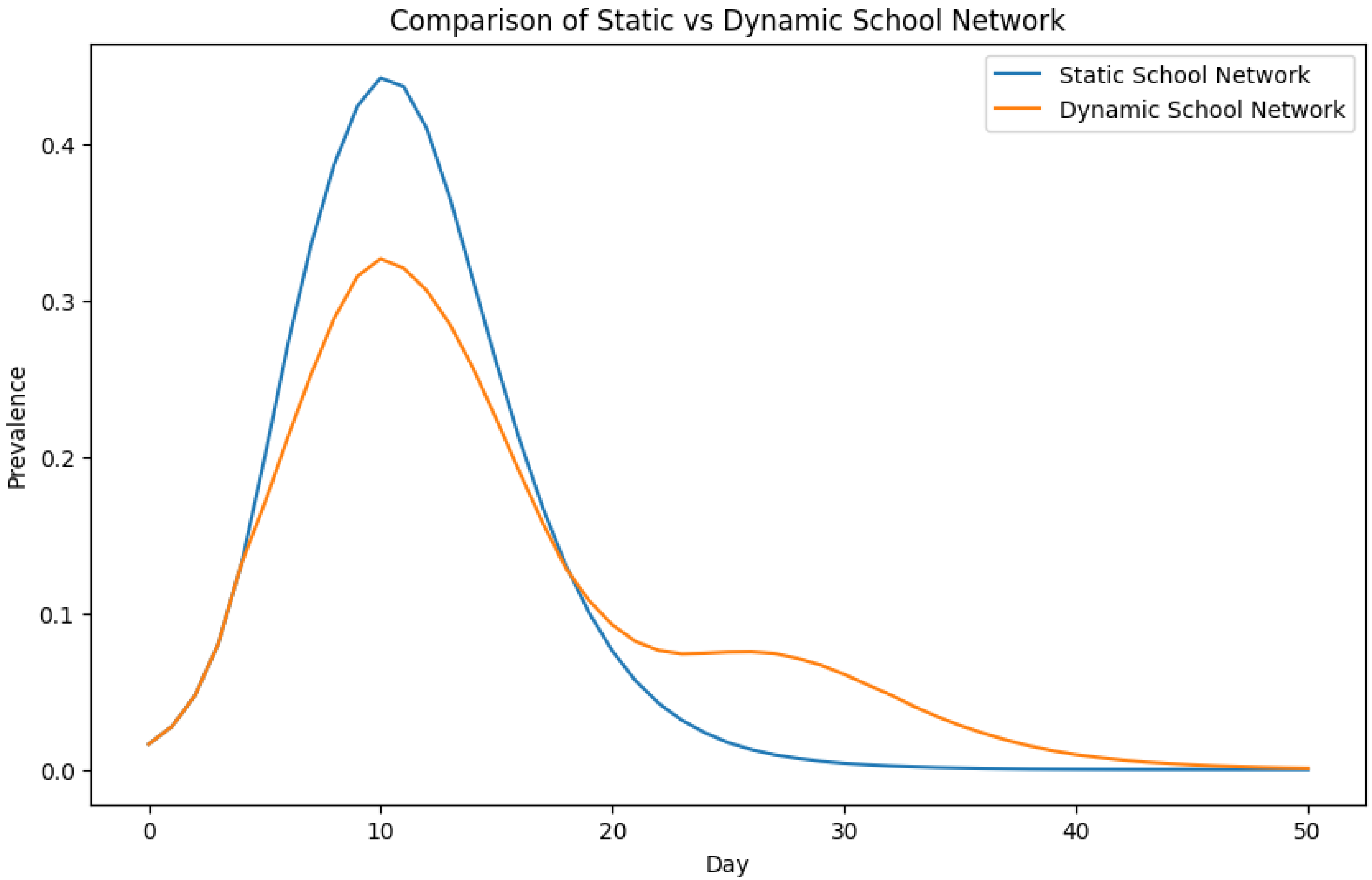
GeoPops agents

Feedback module
- Removes edges of school network when prevalence > 0.1

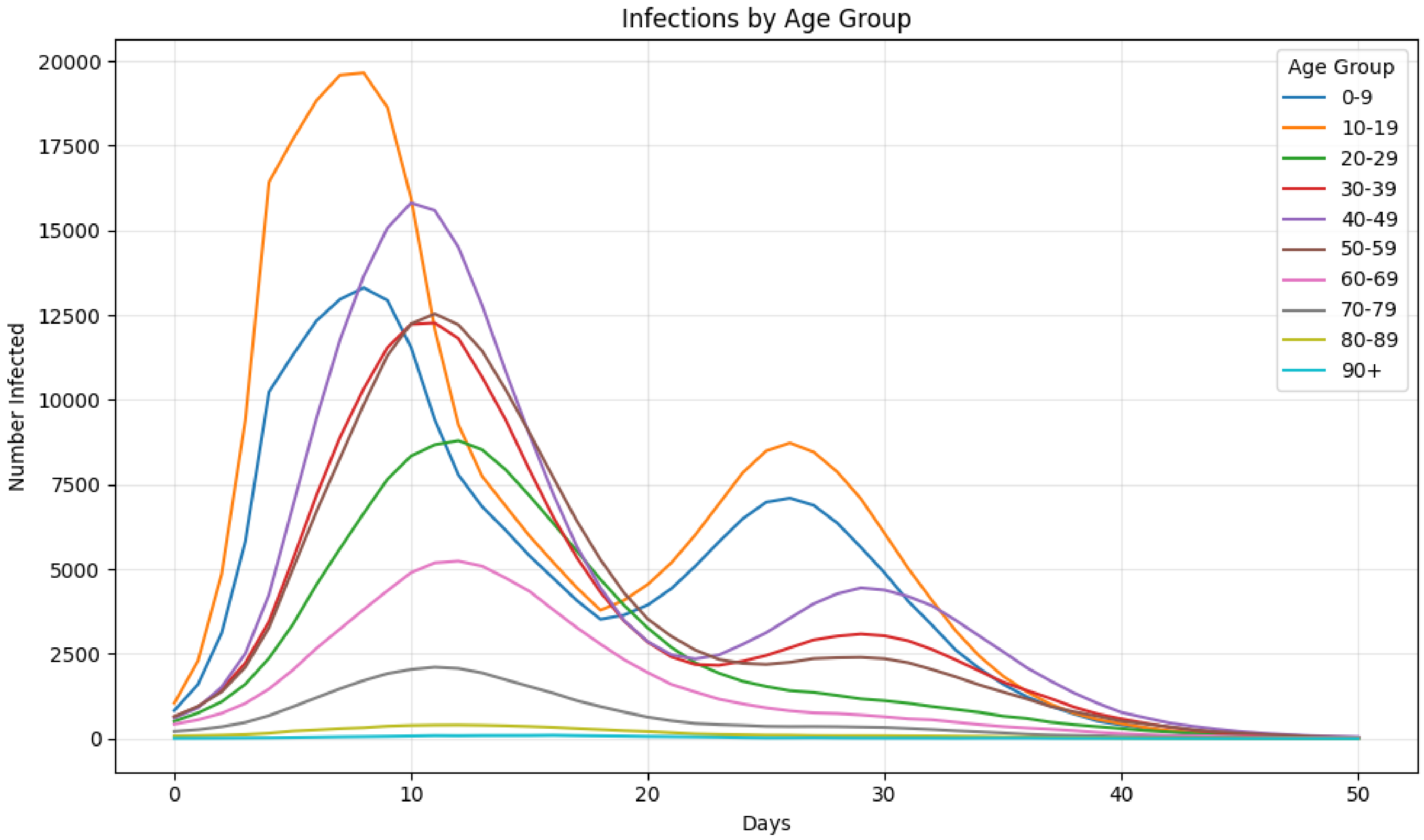
GeoPops networks

Subgroup Tracking
- Infections by age and CBG

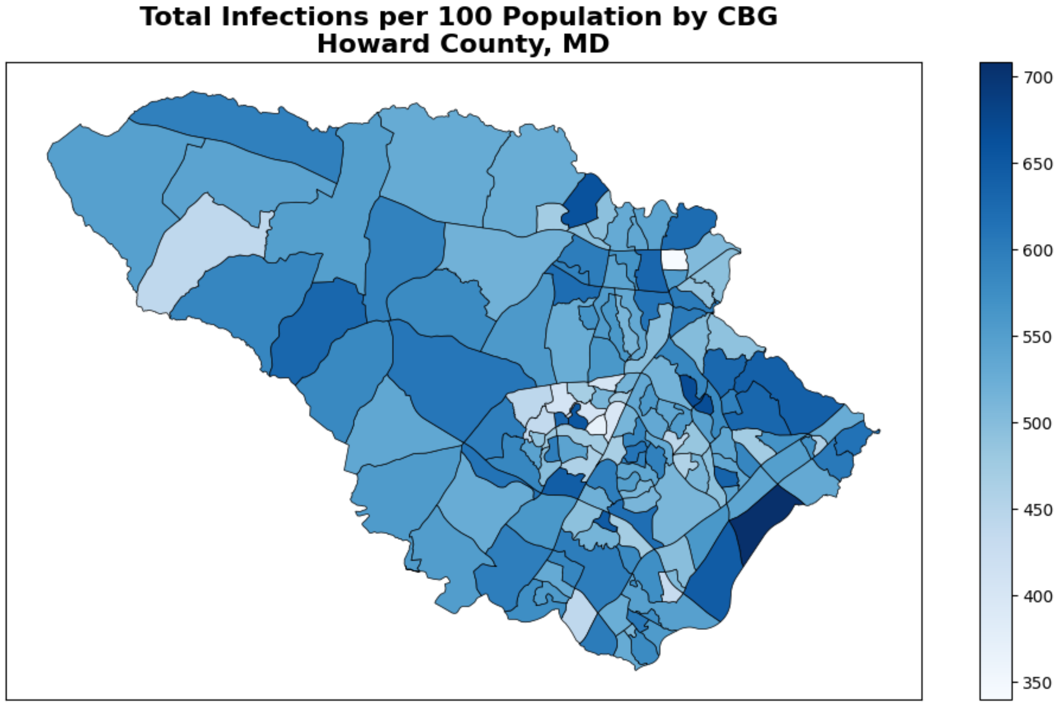
Feedback vs No Feedback

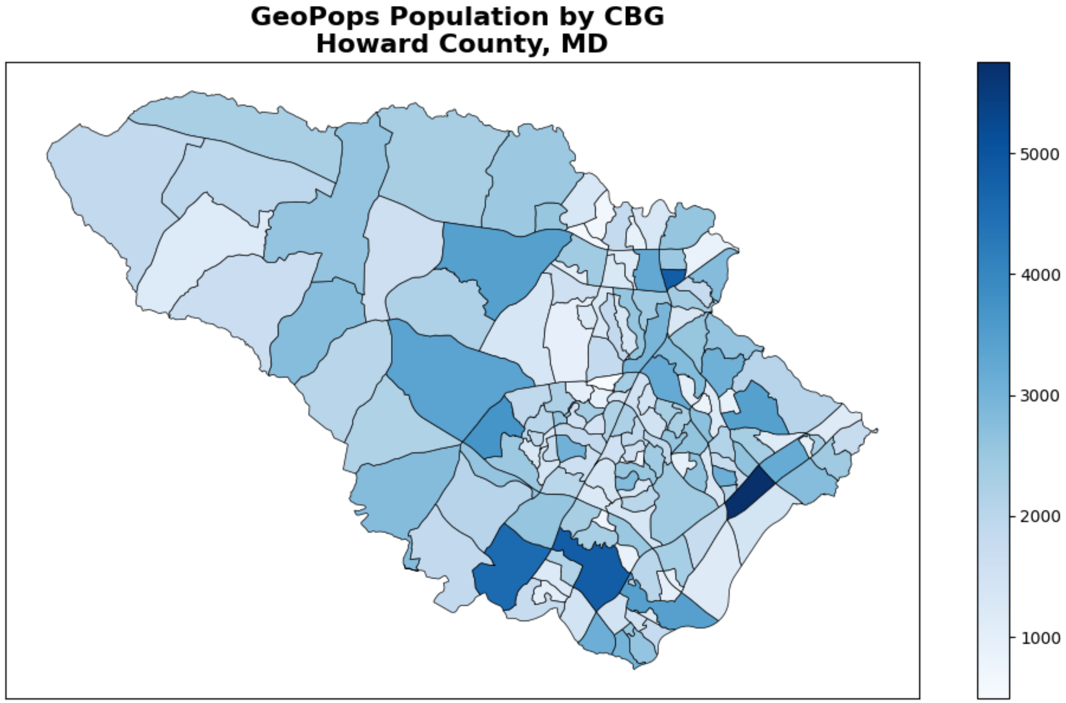


Outcome tracking by demographic subgroup



Outcome tracking by geographic subgroup





Why is this cool?!

- Reasonably realistic geographic and demographic granularity for night and day-time activities
- Don't have to build a model from scratch
- Can be used by state and county health departments for context-specific scenario modeling
- Seed infections to a specific geography
- Target interventions to specific demographic or geographic groups
- There are several research groups already using Starsim to model respiratory diseases that can now use GeoPops as well

Conferences

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Winter Simulation Conference 2025

GeoPops: An open-source package for generating geographically realistic synthetic populations

- More detailed methodology and usage
- Compare GeoPops to two other open-source synthetic population generators (UrbanPop and Geo-Synthetic-Pop)
- Make three MD pops (one with each generator)
- Compare individuals and homes to real Census data
- Compare network stats of each home, school, and workplace network
- Run Starsim COVID-19 model on each population
- Compare simulation results to each other and observed data for first wave of pandemic
- Discuss usability of each generator

Epidemics 2025

Modeling the impact of dynamic decision making on infectious disease outcomes by demographic and geographic subgroups: An open-source agent-based modeling framework

- Run Starsim COVID-19 model with utility framework on GeoPops population of MD
- Individuals weigh health-wealth trade-offs of staying home from work as a function of their income and age
- Test different policy scenarios (e.g., paid sick leave, cash transfer) and compare to observed data from first wave of pandemic
- Demographic and geographic subgroup outcome tracking

Questions

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- Pip install geopops and try going through the tutorial
- Log issues! Please be patient ☺
- Poster, slides, and Notebook tutorial on GitHub

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

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